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Branch: MCA (Data Science)	**Semester:** 2
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Subject Name: Technical Training - I Lab	**Subject Code**: 25CAP-652
Section/Group: 25MCD- KAR1	**Date of Performance:** 31-03-2026

Experiment No.: 8

1. **Aim of the practical:**
   To apply the concept of Stored Procedures in database operations in order to perform tasks like insertion, updating, deletion, and retrieval of data efficiently, securely, and in a reusable manner within the database system.

2. **Tools Used:** PostgreSQL

3. **Objectives of the practical:**
   Apply stored procedure concepts for database operations

4. **Theory:**
   A Stored Procedure is a precompiled collection of SQL statements stored in the database that can be executed whenever required. It is used to perform operations like INSERT, UPDATE, DELETE, and SELECT in a structured and reusable way.
   Stored procedures help in improving performance, security, and code reusability in database systems.
   Key Features:

1. **Precompiled Code**
   Stored procedures are compiled once and stored, which makes execution faster compared to normal SQL queries.
2. **Reusability**
   The same procedure can be called multiple times without rewriting the code.
3. **Security**
   Users can be given permission to execute procedures without direct access to database tables.
4. **Reduced Network Traffic**
   Instead of sending multiple SQL queries, a single procedure call performs multiple operations.
5. **Modularity**
   Complex database logic can be divided into smaller procedures, making it easier to manage.

5. **Experiment / Practical Steps:**

```
CREATE TABLE employee_ (
   emp_id INT  PRIMARY KEY,
   emp_name VARCHAR(100) NOT NULL,
   salary NUMERIC(10,2) NOT NULL,
   department VARCHAR(50)
);

INSERT INTO employee_ (emp_id,emp_name, salary, department) VALUES
(1,'Alice', 50000, 'HR'),
(2,'Bob', 60000, 'IT'),
(3,'Charlie', 55000, 'Finance');

CREATE OR REPLACE PROCEDURE UPDATA_SALARY_PROC(IN P_EMP_ID INT,INOUT
P_SALARY NUMERIC(20,3),OUT
STATUS VARCHAR(20))
   AS
         $$
         DECLARE
         CURR_SAL NUMERIC(20,3);
         BEGIN

        SELECT SALARY+P_SALARY INTO CURR_SAL FROM EMPLOYEE_ WHERE
EMP_ID=P_EMP_ID;
         IF NOT FOUND THEN
         RAISE EXCEPTION 'EMPLOYEE NOT FOUND';
         END IF;
         --CURR_SAL:=CURR_SALARY+P_SALARY;

         UPDATE EMPLOYEE_
         SET SALARY=CURR_SAL
         WHERE EMP_ID=P_EMP_ID;

         P_SALARY:=CURR_SAL;
         STATUS:='SUCCESS';

         EXCEPTION
         WHEN OTHERS THEN
         IF SQLERRM LIKE '%EMPLOYEE NOT FOUND%' THEN
         STATUS:='EMPLOYEE NOT FOUND';
         END IF;
```

```
END
$$ LANGUAGE PLPGSQL;

DO
$$
        DECLARE
        EMP_ID INT:=1;
        STATUS VARCHAR(20);
        SALARY NUMERIC(20,3):=500;
        BEGIN
        CALL UPDATA_SALARY_PROC(EMP_ID,SALARY,STATUS);
        RAISE NOTICE 'YOUR STATUS IS % ',STATUS;
        END;
$$
```

OUTPUT:

```
NOTICE:  YOUR STATUS IS SUCCESS
DO

Query returned successfully in 693 msec.
```

```
DO

 $$
        DECLARE
        EMP_ID INT:=99;
        STATUS VARCHAR(20);
        SALARY NUMERIC(20,3):=500;
        BEGIN
        CALL UPDATA_SALARY_PROC(EMP_ID,SALARY,STATUS);
        RAISE NOTICE 'YOUR STATUS IS % ',STATUS;
        END;
$$
```

OUTPUT:

```
NOTICE:  YOUR STATUS IS EMPLOYEE NOT FOUND
DO

Query returned successfully in 303 msec.
```

SELECT * FROM EMPLOYEE;

OUTPUT:

	emp_id [PK] integer	emp_name character varying (100)	salary numeric (10,2)	department character varying (50)
1	2	Bob	60000.00	IT
2	3	Charlie	55000.00	Finance
3	1	Alice	50500.00	HR

6. **Learning Outcomes:**
   - Understood the concept and importance of stored procedures in database systems.
   - Learned how to create and execute stored procedures for CRUD operations.
   - Gained knowledge about improving database performance and security using procedures.
   - Developed skills to write reusable and modular SQL code for real-world applications.